



Yang Liu

Curriculum Vitae

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Profile

I am a Research Fellow in Maria Secrier's group at University College London. Previously, I was a Postdoctoral Researcher at Hong Kong Baptist University, where I focused on spatial multi-omics data integration, an emerging and highly interdisciplinary field. I hold a PhD in Bayesian network structure learning from Queen Mary University of London, where I developed expertise in probabilistic modelling and machine learning. My research has been published at leading venues, including Nature Communications, ICLR, the Journal of Machine Learning Research, and Machine Learning. At HKBU, I developed disentangled representation learning algorithms to improve the visualisation and annotation of spatial multi-omics data, contributing to advances in this rapidly evolving research area.

Employment experience

- 2026

2027

Research Fellow in computational biology
University of London, UK
- 2024

2026

Postdoc in spatial multi-omics data integration
Hong Kong Baptist University, China

Educational Background

- 2019

2023

PhD in Bayesian network structure learning
Queen Mary, University of London, UK
Thesis: Bayesian network structure learning under imperfect data
- 2017

2018

MSc in Artificial Intelligence
University of Edinburgh, UK
Thesis: Posterior predictive check on Gaussian process
- 2013

2017

BSc in Optoelectronic Information Engineering
Beihang University, China

Publications

- 2026 **Yang Liu**, Kexin Ma, Haoran Xu, Ke Xu, Yunfei Hu, Zhenhan Lin, Jiangli Lin, Bo Han, Shuaicheng Li, Zhixiang Lin, Xin Zhou, and Lu Zhang. Interpretable spatial multi-omics data integration and dimensionality reduction with spamv. *Under review at Nature Communications*, 2026.
- 2025 Zirui Wang, Ke Xu, **Yang Liu**, Yu Xu, and Lu Zhang. Mitigation of multi-scale biases in cell-type deconvolution for spatially resolved transcriptomics using harmodecon. *Bioinformatics*, volume 41, page btaf451. Oxford University Press, 2025.

- 2023 **Yang Liu** and Anthony Constantinou. Improving the imputation of missing data with markov blanket discovery. In *International Conference on Learning Representations*, 2023.
- Neville Kenneth Kitson, Anthony C Constantinou, Zhigao Guo, **Yang Liu**, and Kiattikun Chobtham. A survey of bayesian network structure learning. *Artificial Intelligence Review*, pages 1–94. Springer, 2023.
- Anthony Constantinou, Neville K Kitson, **Yang Liu**, Kiattikun Chobtham, Ar-ian Hashemzadeh Amirkhizi, Prahars A Nanavati, Rendani Mbuva, and Bruno Petrungaro. Open problems in causal structure learning: A case study of covid-19 in the uk. *Expert Systems with Applications*, page 121069. Elsevier, 2023.
- 2022 **Yang Liu**, Anthony C Constantinou, and Zhigao Guo. Improving bayesian network structure learning in the presence of measurement error. *Journal of Machine Learning Research*, volume 23, pages 1–28, 2022.
- Yang Liu** and Anthony C Constantinou. Greedy structure learning from data that contain systematic missing values. *Machine Learning*, volume 111, pages 3867–3896. Springer US, 2022.
- Aiqi Jiang, Xiaohan Yang, **Yang Liu**, and Arkaitz Zubiaga. Swsr: A chinese dataset and lexicon for online sexism detection. *Online Social Networks and Media*, volume 27, page 100182. Elsevier, 2022.
- Anthony C Constantinou, **Yang Liu**, Neville K Kitson, Kiattikun Chobtham, and Zhigao Guo. Effective and efficient structure learning with pruning and model averaging strategies. *International Journal of Approximate Reasoning*, volume 151, pages 292–321. Elsevier, 2022.
- 2021 Anthony C Constantinou, **Yang Liu**, Kiattikun Chobtham, Zhigao Guo, and Neville K Kitson. Large-scale empirical validation of bayesian network structure learning algorithms with noisy data. *International Journal of Approximate Reasoning*, volume 131, pages 151–188. Elsevier, 2021.
- 2020 Anthony C Constantinou, **Yang Liu**, Kiattikun Chobtham, Zhigao Guo, and Neville K Kitson. The bayesys data and bayesian network repository. *Queen Mary University of London: London, UK*, 2020.

Teaching experience

- Demonstrator**
- Bayesian Decision and Risk Analysis, 2020 - 2023
 - Data Analytics, 2022 - 2023

Skills

- R/Matlab/Python
- Pytorch programming experience